

Day 5 Problem Set

Geometry Summer Credit | Triangle Congruence and Corresponding Parts

Name:	Date:	Period:
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Goal

Practice identifying triangle congruence shortcuts and using congruent triangles to find missing sides and angles.

Support Practice

1.	Find the missing triangle angle: 52 degrees, 61 degrees, and x.
2.	Triangle ABC is congruent to triangle DEF. List the matching vertices.
3.	Name the rigid motions from this week.
4.	Is a dilation by scale factor 3 a congruence transformation? Explain.
5.	Solve $4x + 9 = 57$.

Core Practice: Congruence Shortcuts

6.	Two triangles have matching side lengths 4, 6, and 8. Which congruence shortcut applies?
7.	Two triangles have matching sides 5 and 12 with the included angle 40 degrees. Which shortcut applies?

8.	Two triangles have matching angles 35 degrees and 80 degrees with the included side 10. Which shortcut applies?
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9.	Two triangles have matching angles 44 degrees and 61 degrees with a non-included side 7. Which shortcut applies?
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10.	A student says AAA proves triangles are congruent. Explain the mistake.
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11.	A student says SSA always proves triangles are congruent. Explain why that is not a reliable shortcut.
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Day 5 Problem Set

Continued

Core Practice: Missing Parts

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| 12. | Triangle ABC is congruent to triangle RST. If $AB = 13$, $BC = 9$, and angle $A = 48$ degrees, find RS, ST, and angle R. |
| 13. | Triangle JKL is congruent to triangle MNP. If angle $K = 72$ degrees, which angle equals 72 degrees? |
| 14. | Triangle DEF is congruent to triangle XYZ. If $EF = 3x - 2$ and $YZ = 19$, find x . |
| 15. | Triangle GHI is congruent to triangle QRS. If angle $H = 2x + 15$ and angle $R = 79$ degrees, find x . |
| 16. | Triangle ABC is congruent to triangle DEF. If $AB = 2x + 5$ and $DE = 21$, find x and AB. |
| 17. | Triangle PQR is congruent to triangle LMN. If angle $R = 4x - 6$ and angle $N = 98$ degrees, find x . |
| 18. | Error analysis: A student sees triangle ABC congruent to triangle DEF and says angle A matches angle F. Explain the error. |
| 19. | Spiral: Rotate $(-7, 2)$ 90 degrees counterclockwise about the origin. |
| 20. | Spiral: Lines have slopes $\frac{3}{4}$ and $-\frac{4}{3}$. Are they parallel, perpendicular, or neither? |

21.	Challenge: Explain how triangle congruence connects to rigid motions from Day 4.
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Before You Submit

Star two problems: one shortcut problem and one missing-parts problem.